

# ME2 - Voice Controlled Smart Device

## Always-on wake word + intent recognition at the edge

3 October 2026 | Live acoustic benchmark | Raspberry Pi 5 | 125 trials

### Model and runtime

Mic 16 kHz -> 40-bin log-mel features

Wake gate -> 31-class INT8 intent model

Intent + slot -> local Raspberry Pi action

| Model item     | Value                    |
|----------------|--------------------------|
| Intent weights | 15,971 params   37.5 KB  |
| Inference      | 7.9 ms mean   9.3 ms p95 |

### Live Pi 5 result

| Metric                     | Measured         |
|----------------------------|------------------|
| Intent accuracy            | 36.7% (40 / 109) |
| Command accuracy           | 33.9%            |
| Out-of-scope false accepts | 62.5% (10 / 16)  |
| No-wake false fires        | 50.0% (8 / 16)   |

### Evaluation set

| Item                 | Value                  |
|----------------------|------------------------|
| Wake-positive trials | 109                    |
| In-scope phrases     | 93 command variants    |
| Out-of-scope clips   | 16                     |
| No-wake controls     | 16                     |
| Voice mix (scored)   | 49 real   60 synthetic |

### Raspberry Pi 5

| Runtime item    | Value                  |
|-----------------|------------------------|
| Platform        | 4-core   4 GB RAM      |
| Inference RTF   | 0.003 mean   0.004 p95 |
| CPU temperature | 45.2 C maximum         |
| Throttling      | None observed          |

Quick holdout benchmark; this evaluates the existing deployed ME2 model, not a retrained candidate.

# Live acoustic benchmark

| Metric                     | Overall | Real voice | Synthetic voice |
|----------------------------|---------|------------|-----------------|
| Intent accuracy            | 36.7%   | 16.3%      | 53.3%           |
| Command accuracy           | 33.9%   | 12.2%      | 51.7%           |
| Intent macro F1            | 34.7%   | 12.0%      | 49.9%           |
| Out-of-scope false accepts | 10 / 16 | Conflicted | Conflicted      |

## Pi inference

| Measure              | Result                 |
|----------------------|------------------------|
| Intent inference     | 7.9 ms mean            |
| Intent inference p95 | 9.3 ms                 |
| Real-time factor     | 0.003 mean / 0.004 p95 |
| Peak CPU temperature | 45.2 C                 |

## Interpretation

### Recognition is the bottleneck

The model runs quickly on the Pi, but real-voice intent accuracy is only 16.3%. Overall false accepts were 10 of 16 out-of-scope clips.

**The test logged 40 no-responses and 49 extra fires across 109 wake-positive trials.**

## Review before making model changes

- 1 Audit the acoustic event log: 16 of 69 response-latency samples are negative, so the reported 6.99 s p95 is provisional.
- 2 Repeat with a larger, balanced holdout and corrected event-to-utterance timing before claiming end-to-end latency or promoting a model.

125 trials total. FAR source breakdown conflicts in report prose; no retraining or promotion was performed.